Yuval Kansal

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EDUCATION

Princeton University *Princeton, USA*
PhD in Electrical and Computer Engineering **2026-Present**
M.A. in Electrical and Computer Engineering (PROGRAM GPA: 3.96/4) **2024-2026**

- Advised by Prof. Niraj Jha (Specialization: Reasoning, Reliability in LLMs; Domain-specific Superintelligence)
- Interested in superintelligence and building frontier AI systems capable of composing from first principles.
- Recipient of the Hisashi and Masae Kobayashi *67 Fellowship
- Relevant Coursework: NLP, RL, Machine Learning, Smart Healthcare

University of New South Wales (UNSW Sydney) **June 2020 – May 2024**
Bachelor of Engineering (Honors), Computer Engineering *Sydney, Australia*

- Completed my Honors Research Thesis entitled "Improving Radio Frequency Fingerprinting by leveraging denoising Generative models," advised by Prof. Salil Kanhere and Dr. David Nguyen, with High Distinction WAM (Honors Class 1).
- Recipient of International Scientia Coursework Scholarship (100% tuition fee waiver) for the entire duration of my studies.
- Part of Dean's Honors List 2020 and 2021 (Highly Commended), Winner of the Circular Economy Hackathon 2020, Google GDSC Campus Lead 2021-22, UNSW Student Ambassador since 2021.

RESEARCH WORK EXPERIENCE

FAIR, Meta Superintelligence Labs (MSL) **May 2026 – August 2026**
Research Scientist Intern *New York City, USA*

- Research Scientist Intern in the AI Alignment Team. Manager: Mark Ibrahim
- Built Self-Direction, which turns agent failures into reusable LoRA lessons so agents fix their own mistakes and use them across retries and related tasks, enabling continual improvement.

CSIRO **April 2024 – July 2024**
PhD Scholarship *Sydney, Australia*

- Research intern at Data61 in CSIRO for 14 weeks.
- Worked on the project – "Software secure code generation using Large Language Models."
  - Advisors: Dr. Sharif Abuadbba and Dr. Kristen Moore
- Created a pipeline to automatically verify and fix security vulnerabilities present in LLM-generated code without a human in the loop.

UNSW School of CSE **February 2024 – July 2024**
Research Officer *Sydney, Australia*

- Research Officer under Dr. Jake Renzella and Dr. Sasha Vassar on the project "Integrating Large Language Models into the Debugging C Compiler for Generating Contextual Error Explanations".
- Created a fine-tuned agent on a custom dataset to improve LLM output in line with the pedagogical standards and conducted extensive user studies.

Canopus Networks, UNSW **February 2022 – August 2022**
Data Analytics Intern *Sydney, Australia*

- Awarded the Taste of Research Scholarship to pursue a 60-day internship under Prof. Vijay Sivaraman.
- Analyzed gaming UX data of 1,000,000+ records at a time from servers across Australia and produced quarterly reports published on the company's website.

University of Fribourg/ Swiss Embassy **September 2021 – November 2021**
Research Intern *Fribourg, Switzerland*

- Awarded the prestigious ThinkSwiss Research Scholarship to pursue a 3-month internship at the University of Fribourg, Switzerland.
- Part of the DIVA group under Prof Ingold, where I worked on a model for semantic segmentation on a synthetically generated historical dataset and aligned it with a framework that was built by other senior students.

SELECTED PUBLICATIONS (See Google Scholar for more details)

- **Don't Forget What You Just Learned: Self-Directed Repair Turns Agent Mistakes Into Lasting Lessons** (arXiv preprint)
  ***Yuval Kansal**, Karen Ullrich, Polina Kirichenko, Ilia Kulikov, Niraj K. Jha, Mohammad Pezeshki, Mark Ibrahim*

- **An Alternative Trajectory for Generative AI** (arXiv preprint)
  Margarita Belova, **Yuval Kansal***, Yihao Liang*, Jiaxin Xiao*, Niraj K. Jha*

- **Knowledge Graphs are Implicit Reward Models: Path-Derived Signals Enable Compositional Reasoning** (arXiv preprint)
  ***Yuval Kansal**, Niraj K. Jha*

- **Bottom-up Domain-specific Superintelligence: A Reliable Knowledge Graph is What We Need** (arXiv preprint)
  *Bhishma Dedhia, **Yuval Kansal**, Niraj K. Jha*

- **Facts Do Care About Your Language: Assessing Answer Quality of Multilingual LLMs,** (*AAAI 2025 iRAISE Workshop)*
  Shmuel Berman, **Yuval Kansal***, Lydia Liu*

- **From Solitary Directives to Interactive Encouragement! LLM Secure Code Generation by Natural Language Prompting** *(arXiv preprint)*
  Shigang Liu, Bushra Sabir*, Seung Ick Jang, **Yuval Kansal**, Yansong Gao, Kristen Moore, Alsharif Abuadbba, Surya Nepal*

- **GuideLM: A Pedagogically-Aligned LLM for Programming Education** *(AAEE 2026)*
  Emily Ross, **Yuval Kansal***, Jake Renzella, Alexandra Vassar, Andrew Taylor*

PATENTS

- Bottom-up Domain-specific Superintelligence: A Reliable knowledge graph is what we need **(Provisional)**
  INVENTORS: Bhishma Dedhia, Yuval Kansal, Niraj Jha

- Knowledge graphs are implicit reward models: Path-Derived signals enable composition reasoning **(Provisional)**
  INVENTORS: Yuval Kansal, Niraj Jha

Invited Talks

- **Google Research NYC – Building Human-like Superintelligence in AI Models** (March 2026)
- **Google DeepMind India – Building Human-like Superintelligence in AI Models** (February 2026)
- **Ploutos Dev – Knowledge Graphs are Implicit Reward Models** (February 2026)
- **AI Lab Princeton – Bottom-up Domain-specific Superintelligence** (December 2025)

SERVICE

- ICML 2026 Reviewer (GOLD, top 25%, 6 papers), AAAI 2026 Reviewer (5 papers)
- LLM Cyber-Sec Workshop at IEEE TPS 2024 (Organizer)
- Classroom Champion cum Volunteer for Day of AI Australia 2024.